



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

2.0 Introduction to Industrial Internet of Things (IIoT)

July 2025

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Records of Change

Date of Issue/ Reissue	Revision No.	Reason of Development / Change
July 2025	00	New module offering inline with TI&R 2025.

SKG14 TI&R Recap

7-3-13-0003-QR2 v3.0 SKG 14 – BASE Technology Competency Matrix / Descriptor Discipline – Instrument & Control Sub-Discipline – Instrument & Control Effective Date – 01 July 2025						
TECHNOLOGY INVENTORY		COMPETENCE LEVELS				
		Awareness	Knowledge	Skill	Advanced	Expert
P1	Industrial Internet of Things  The use of Industrial Internet of Things (IIoT) technologies by utilising the capabilities of physical devices embedded with electronics, sensors, actuators, software and network (emphasis on wireless connectivity). This enables the devices to connect seamlessly with each other for data exchange, information processing and Edge/Cloud computing to increase performance in efficiency, productivity and/or reliability.  Digitalization	Aware of the difference between IIoT vs IoT. Describe IIoT components and architecture.	Provide example of IIoT implementation used in oil and gas industry to improve plant and equipment performance. Know security aspects that should be included in order to improve data exchange.	Involve in IIoT implementation and demonstrate: - Architecture of implementation - Communication standards and wireless communication protocols used. - Key components and cybersecurity features of the architecture	Justify the IIoT implementation features selected with reference to both operational and economic factors. Mentoring, coaching or training (within OPU or SKG).	



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Learning Outcome

At the end of the session, you will be able to:

- Explain the significance of IoT and its transformative impact on industries and daily life.
- Understand the core concepts of IoT and its applications in the Industrial, Commercial, and Consumer domains.
- Describe the foundational principles and technologies that underpin IIoT systems.
- Understand how sensors and actuators enable data collection and control in industrial settings.
- Explore computing paradigms like Cloud, Fog, and Edge computing and their role in IIoT.
- Explain the communication methods, standards, and layers that facilitate connectivity in IIoT systems.

02.0 Introduction to IIoT

- Why IoT?
- What is IoT?
- Basics of Industrial Internet of Things (IIoT)
- Industrial Sensing & Actuation
- IIoT Compute
- IIoT Connectivity

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Why IoT?

- Why IOT is crucial to our business?
- Enhanced Business Efficiency
- Market Leadership
- How is our business is positively affected by IOT ?
- Among positive effects of utilizing IOT in our business are :
 1. New revenue streams
 2. Operational efficiencies
 3. Improved decision-making
 4. Enhanced customer experiences
 5. Enhancement of existing business models
 6. New partnerships and collaborations
 7. Improved safety



Industrial Internet of Things (IIoT)

Self-Reading

The benefits of IIoT drivers are



Improved operational efficiency of the different industrial processes



Reduced product development and assembly line-related costs (operational cost)



Machine downtime is reduced, and energy is conserved



Predictive and prescriptive analysis of machines using their data



Remote monitoring of the factory's operations



Improved estimation of available materials, work progress, and the arrival of inventory



Safety and security of workers is enhanced



Better customer experience and product packaging.



Creation of new business opportunities.

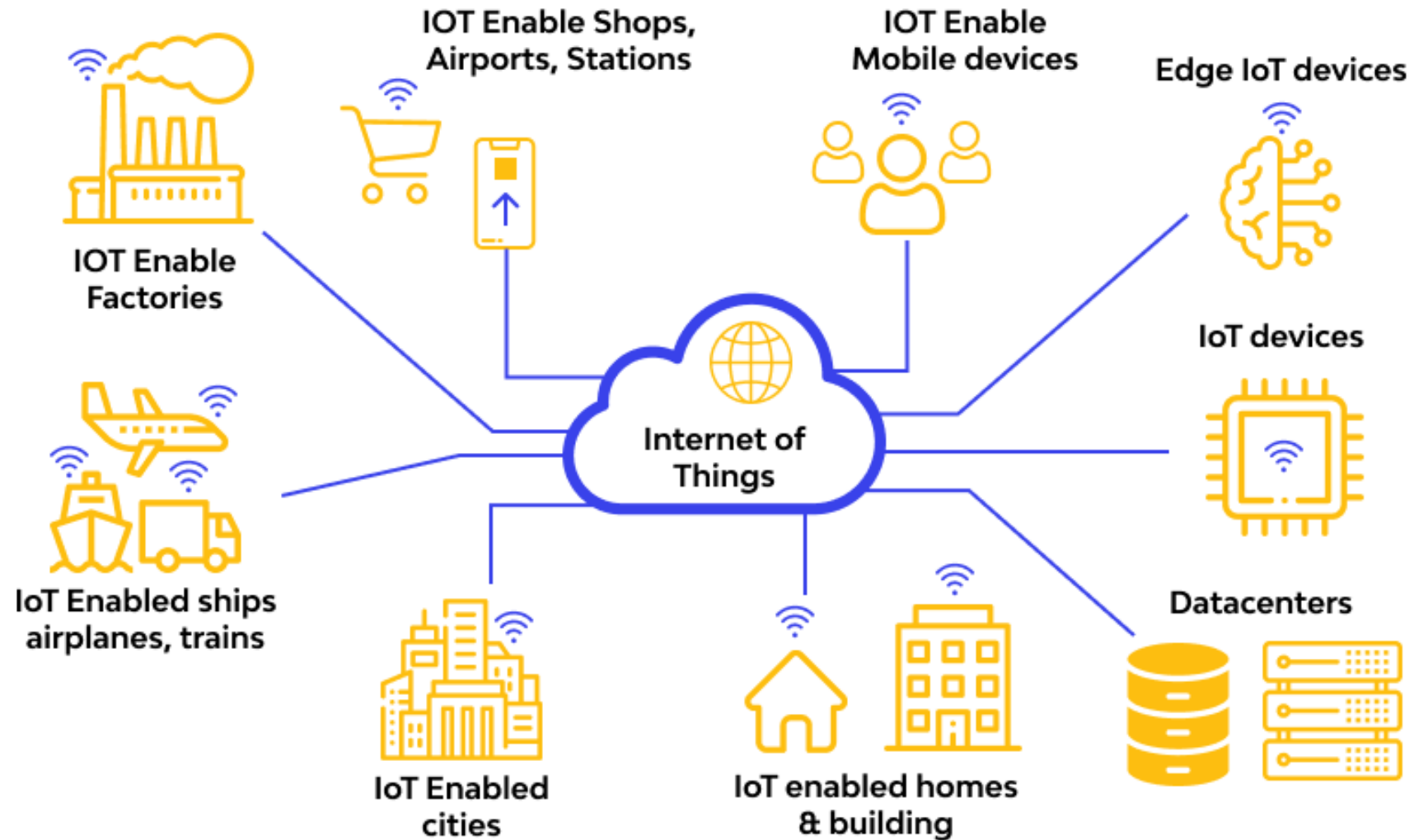


The real-time supply chain monitoring.

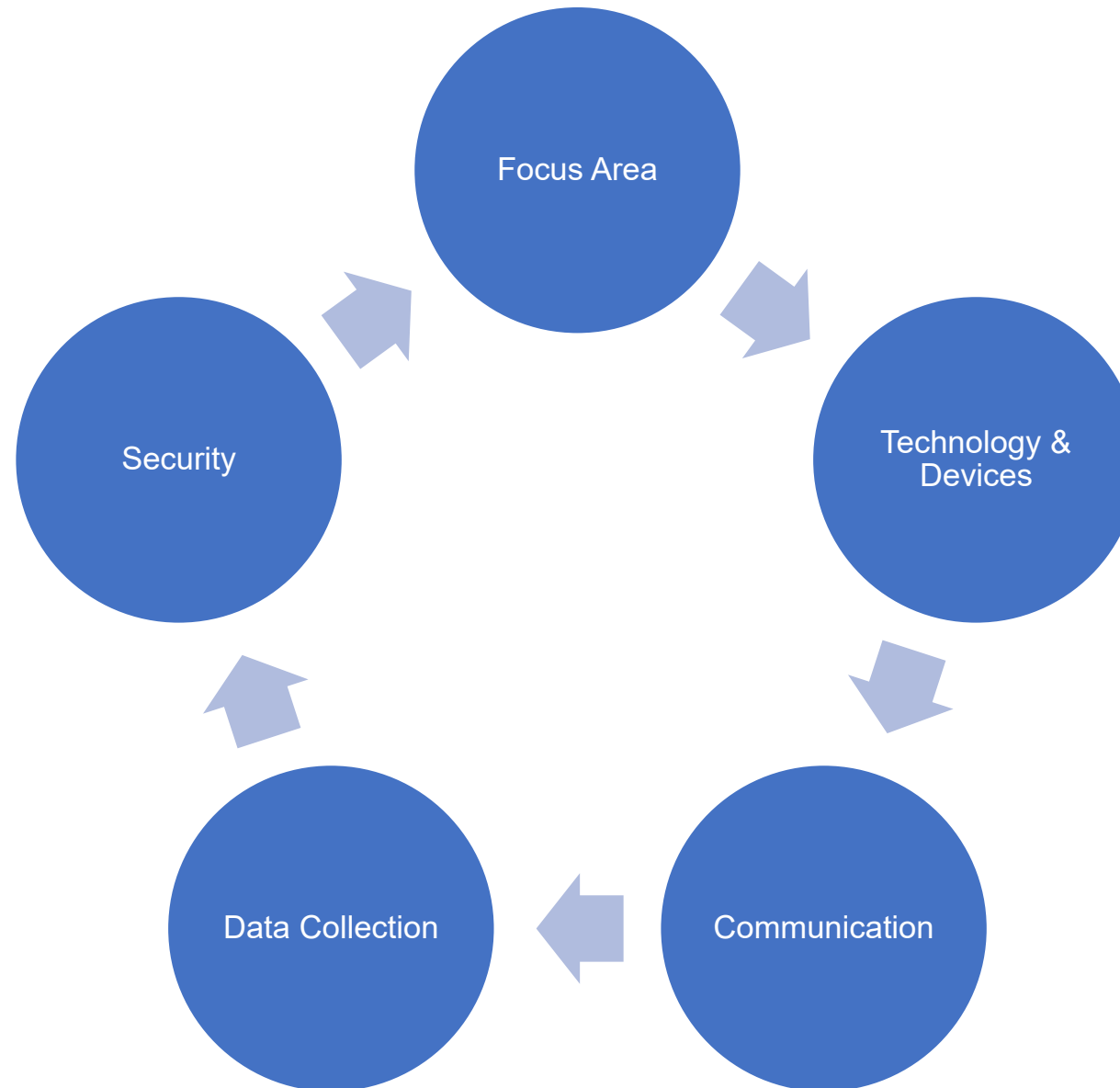
02.0 Introduction to IIoT

- Why IoT?
- **What is IoT?**
- Basics of Industrial Internet of Things (IIoT)
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- IIoT Connectivity

What is IoT?



How IoT differs from OT ?



What is IoT?

Industrial IoT (IIoT)

- IIoT integrates IoT technology in industrial settings, where machines, sensors, and devices are connected to the internet to collect, exchange, and analyze data.
- IIoT enhances manufacturing processes, supply chains, energy management, and maintenance systems by enabling real-time monitoring, automation, and decision-making.
- IIoT devices such as sensors, actuators, and control systems are embedded in industrial machinery like turbines, assembly lines, and pumps, gathering critical data on performance, health, and operational conditions.
- The data is transmitted to centralized platforms or cloud-based systems where it is analyzed to provide actionable insights, predictive maintenance, and process optimization.

Difference between IoT and IIoT



What is IoT?

Industrial IoT (IIoT)

Key Differences Between IoT and IIoT :

IoT vs. IIoT

Features	IOT	IIOT
Primary Application	Consumer applications (homes, wearables, etc.) Devices are developed for the well-being of individuals.	Industrial and enterprise applications (factories, power plants, etc.) Devices are developed for the improvement of efficiency, safety and security in industries
Devices	Everyday objects like smartphones, smart home devices	Industrial machines, sensors, control systems, robotics
Data Volume	Smaller amounts of data, often less complex	Large volumes of high-frequency, complex data from machinery
Connectivity	M-2-M communication possible in small scale via Wi-Fi, Bluetooth, cellular, etc.	Ethernet, industrial protocols (Modbus, OPC UA, etc.) M2M communication, SCADA, and other manufacturing technologies are applicable in large scale
Reliability & Security	Moderate reliability and security (consumer-grade) with limited or no human involvement.	High reliability, security, and uptime are critical due to operational impact which require control and automation logic, deep learning, and analytics
Real-Time Operations	Less emphasis on real-time operations	Real-time data processing and decision-making are essential

What is IoT?

Commercial IoT

- Utilized in businesses and industries to improve operations, productivity, and decision-making.
- Interconnection between devices, sensors, and systems for data gathering , performance monitoring, and process automation.
- Commercial IoT business applications : manufacturing, logistics, retail, healthcare, agriculture etc.
- Key areas involving Commercial IoT



Manufacturing



Retail



Supply chain & logistics



Smart Buildings



Agriculture



Healthcare

What is IoT?

Consumer IoT

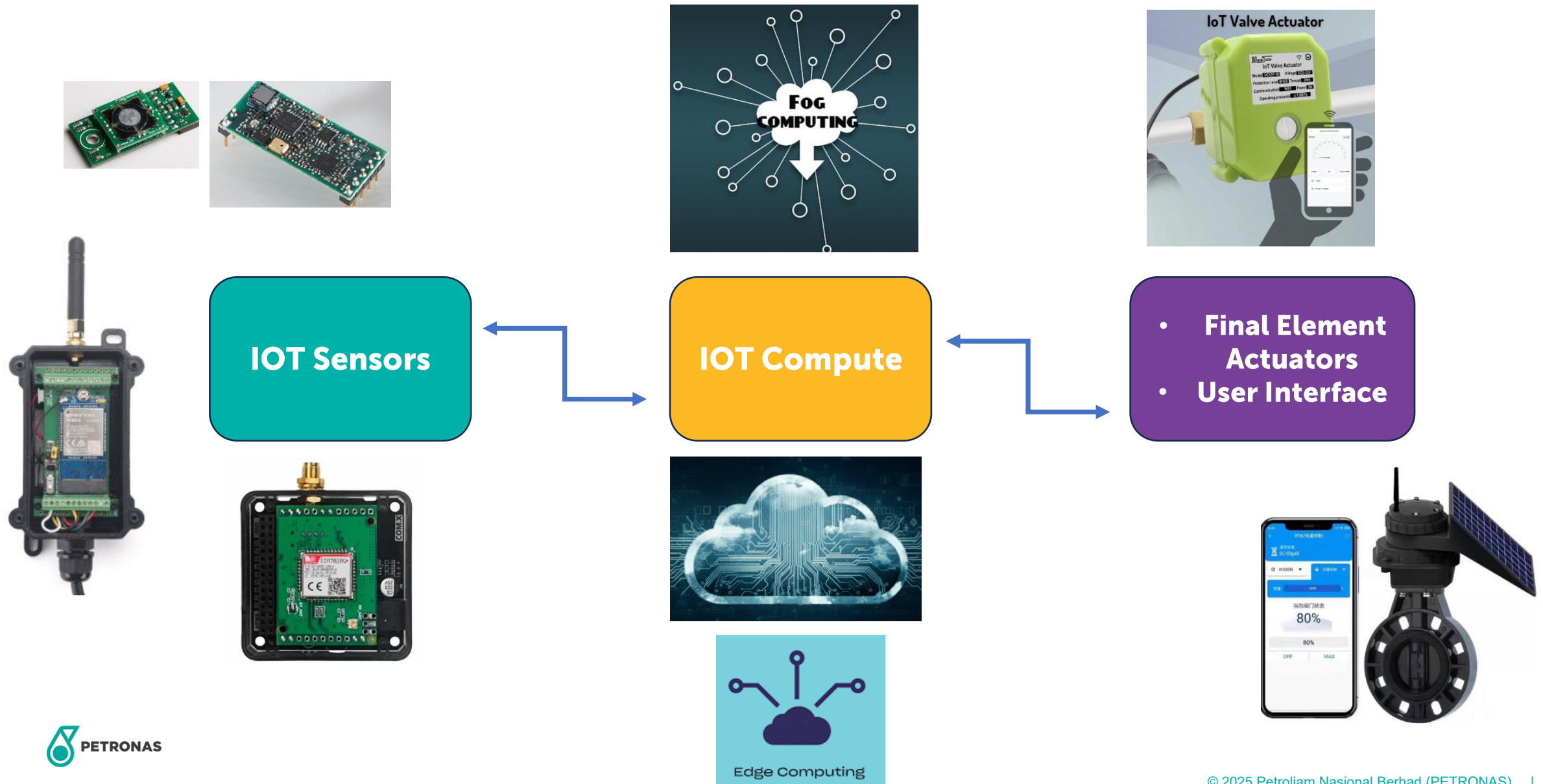
- Consumer IoT is a network of smart devices used daily to enhance convenience, efficiency, and connectivity. These devices are connected to the internet and can interact with each other, often controlled via mobile apps, voice assistants, or automated systems.
- Key features of consumer IOT includes :
 - **Remote Monitoring and Control:** Users can access and manage their devices from anywhere.
 - **Automation:** Devices can be set to automatically perform tasks based on schedules or triggers (e.g., lights turning on at sunset).
 - **Data Collection and Analytics:** Many IoT devices collect data for insights into user behavior, preferences, and device performance.
 - **Voice and App Control:** Integration with mobile apps and virtual assistants like Alexa, Google Assistant, or Siri for voice commands.
- Examples of consumer IOT system :
 - **Smart Home Devices :** Nest , Alexa , Google Home
 - **Wearables :** Apple Watch, Fitbit
 - **Connected Appliances:** Refrigerators, Washing machines
 - **Connected Vehicles**
 - **Smart Entertainment**



02.0 Introduction to IIoT

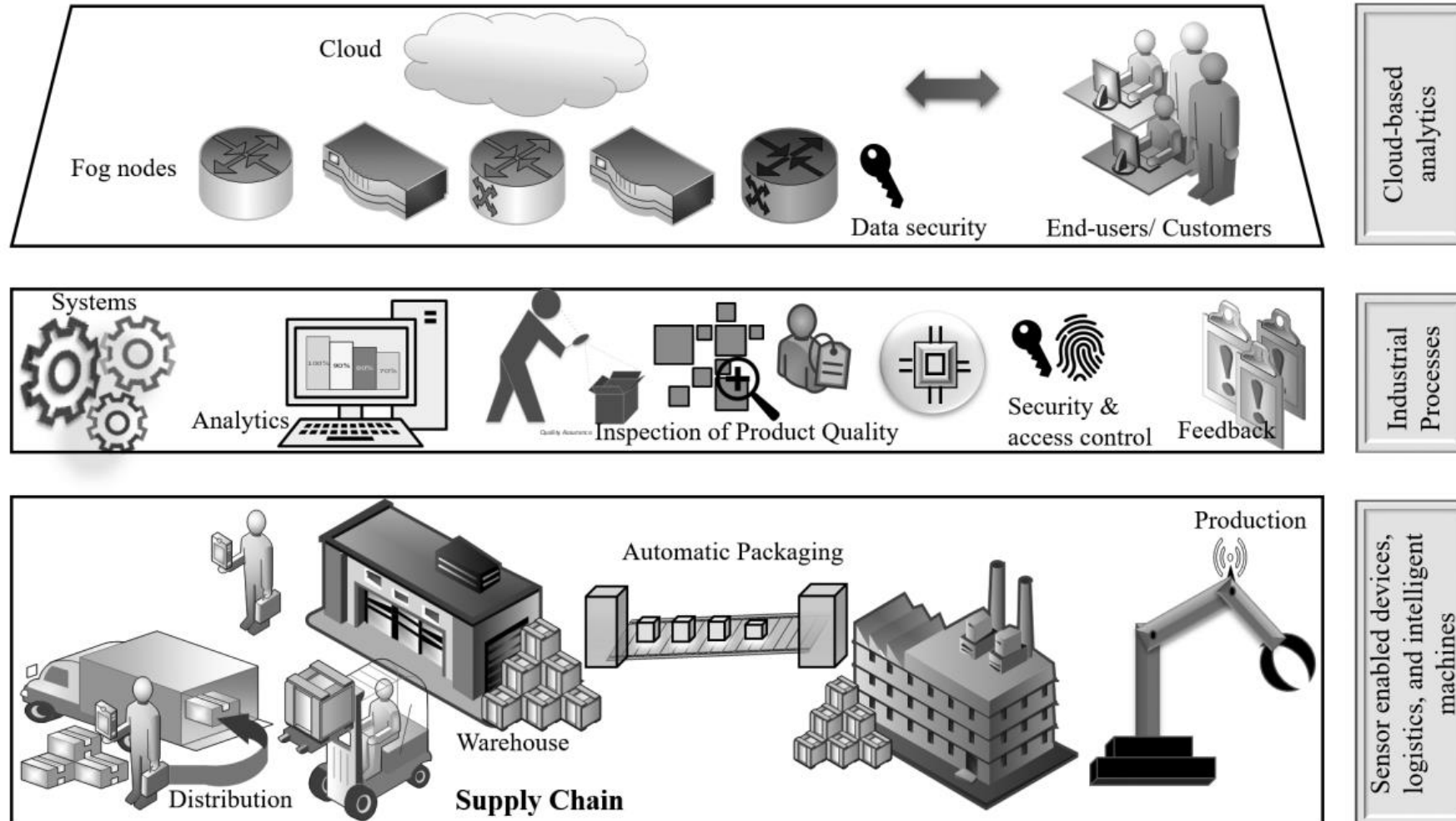
- Why IoT?
- What is IoT?
- **Basics of Industrial Internet of Things (IIoT)**
- Industrial Sensing & Actuation
- IIoT Compute
- IIoT Connectivity

Basics of Industrial IoT: Industrial Processes & Internet Systems



IIoT: Industrial Processes

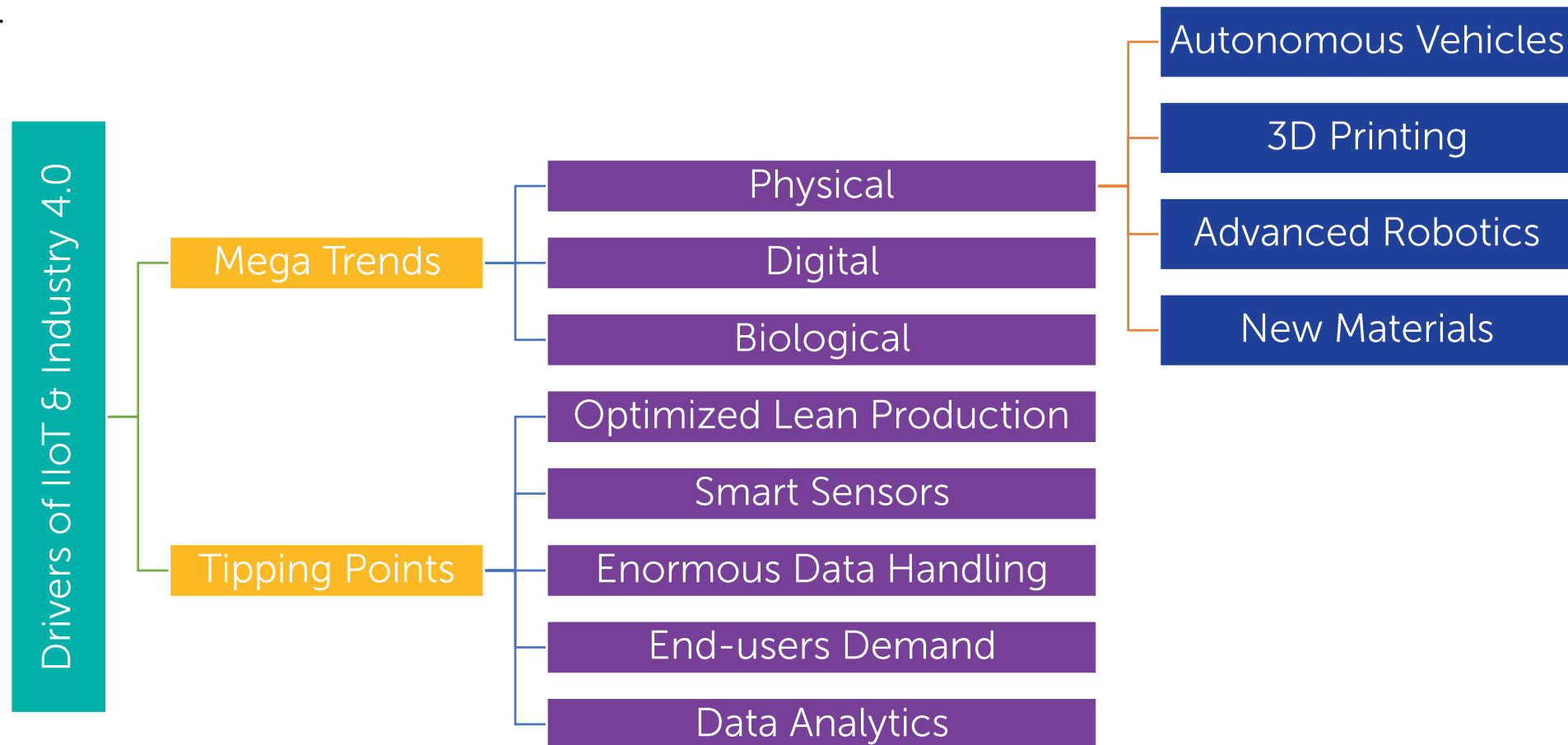
Self-Reading



Basic of Industrial IoT

Drivers of IIoT & Industry 4.0

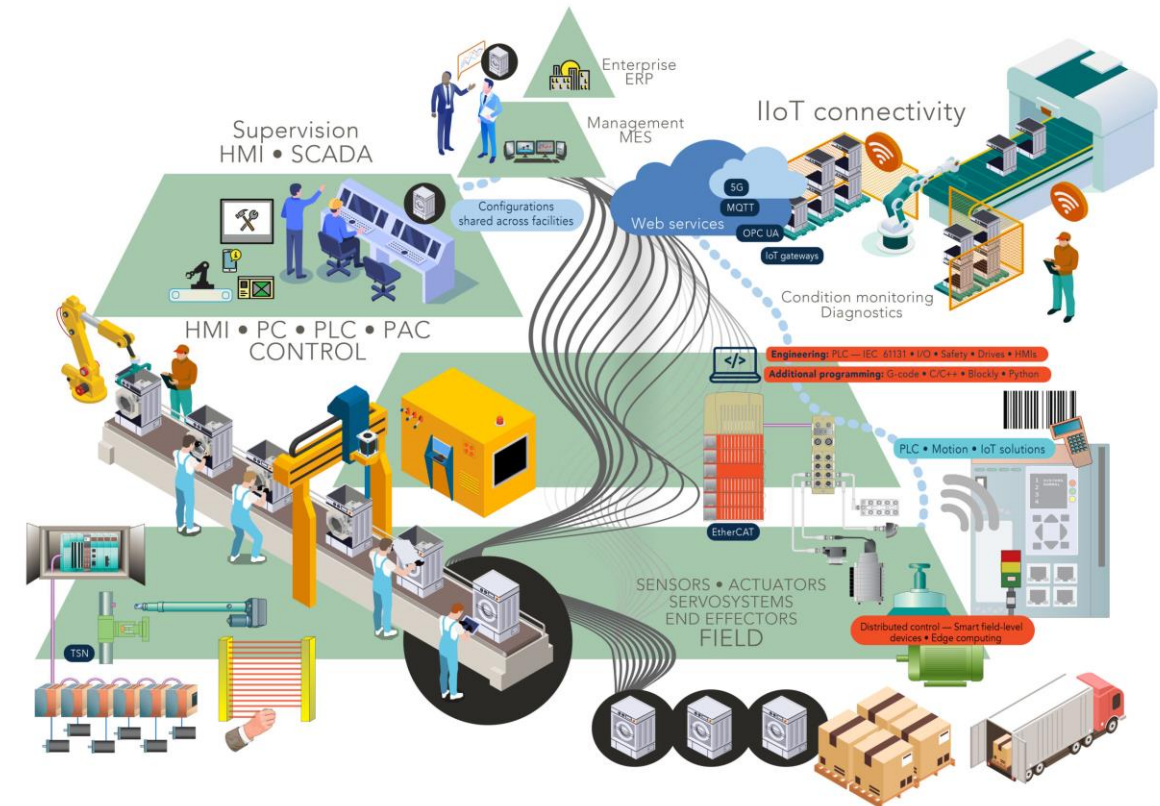
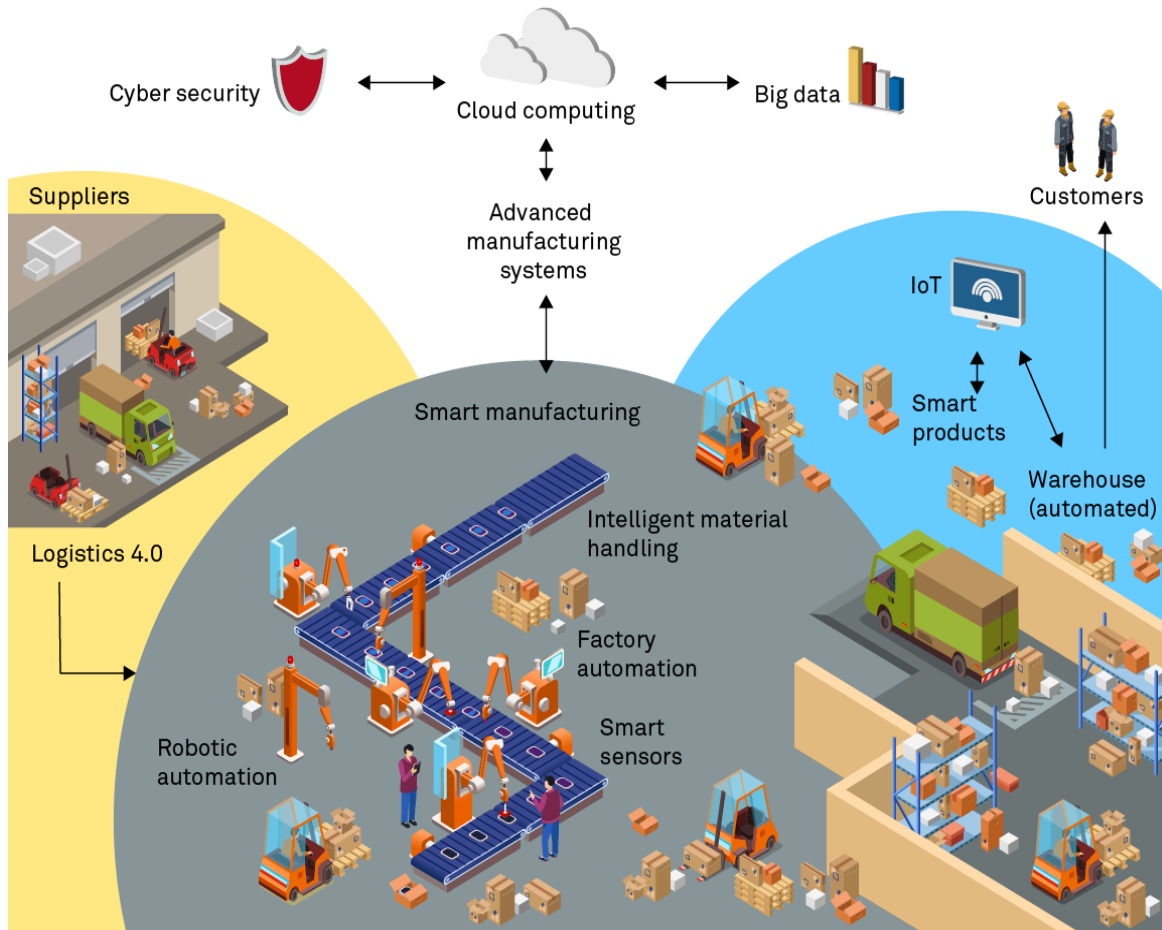
- The industrial revolutions over the years have brought considerable technological changes to the industries.
- The various Internet-based advanced technologies have converted industrial systems and processes into smarter ones.



Basic of Industrial IoT

Industrial Internet Systems

The term Industrial Internet is coined by General Electric (GE), which combines physical and digital worlds.



Source: <https://www.designworldonline.com/smart-factories-and-machine-connectivity/>; <https://www.designworldonline.com/smart-factories-and-machine-connectivity/>

Basic of Industrial IoT

Industrial Internet Systems

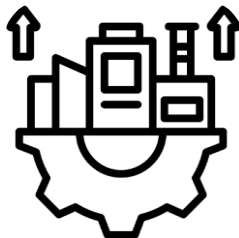
The catalysts behind the development of the Industrial Internet are



The investment in the deployment of sensor nodes, required instruments, business processes, and updated user application interfaces will promote the development of new technologies.



Securing sensitive information, data, and intellectual property (IP) requires a robust security system.

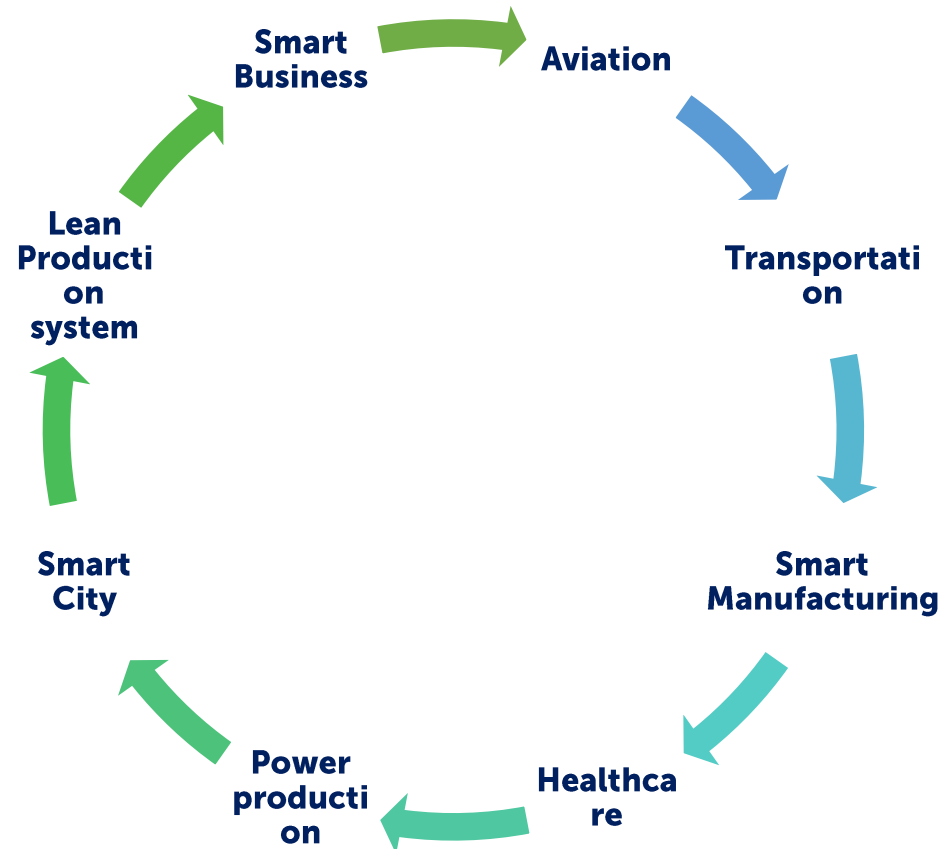


To encourage new talents such as advanced engineering studies, data scientists, and software experts to promote industrial development.

Applications of Industrial Internet in Industry 4.0

Self-Reading

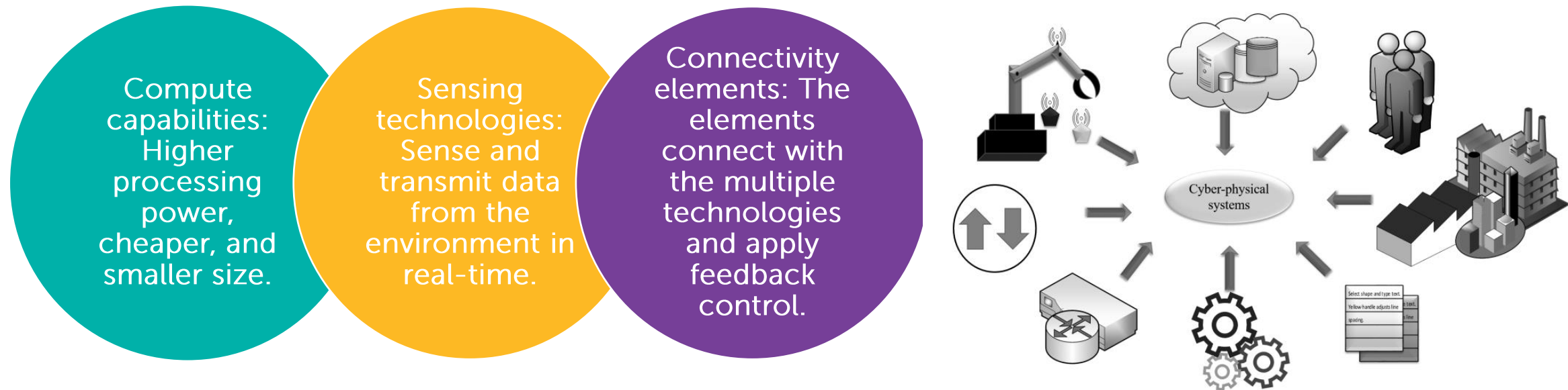
- The Industrial Internet has transformed the industrial sector, with the help of available resources such as smart sensor nodes, software, and M2M communication.
- The product's efficiency improves in time and cost, quality, and overall productivity.
- It has a wide range of applications



Basic of Industrial IoT

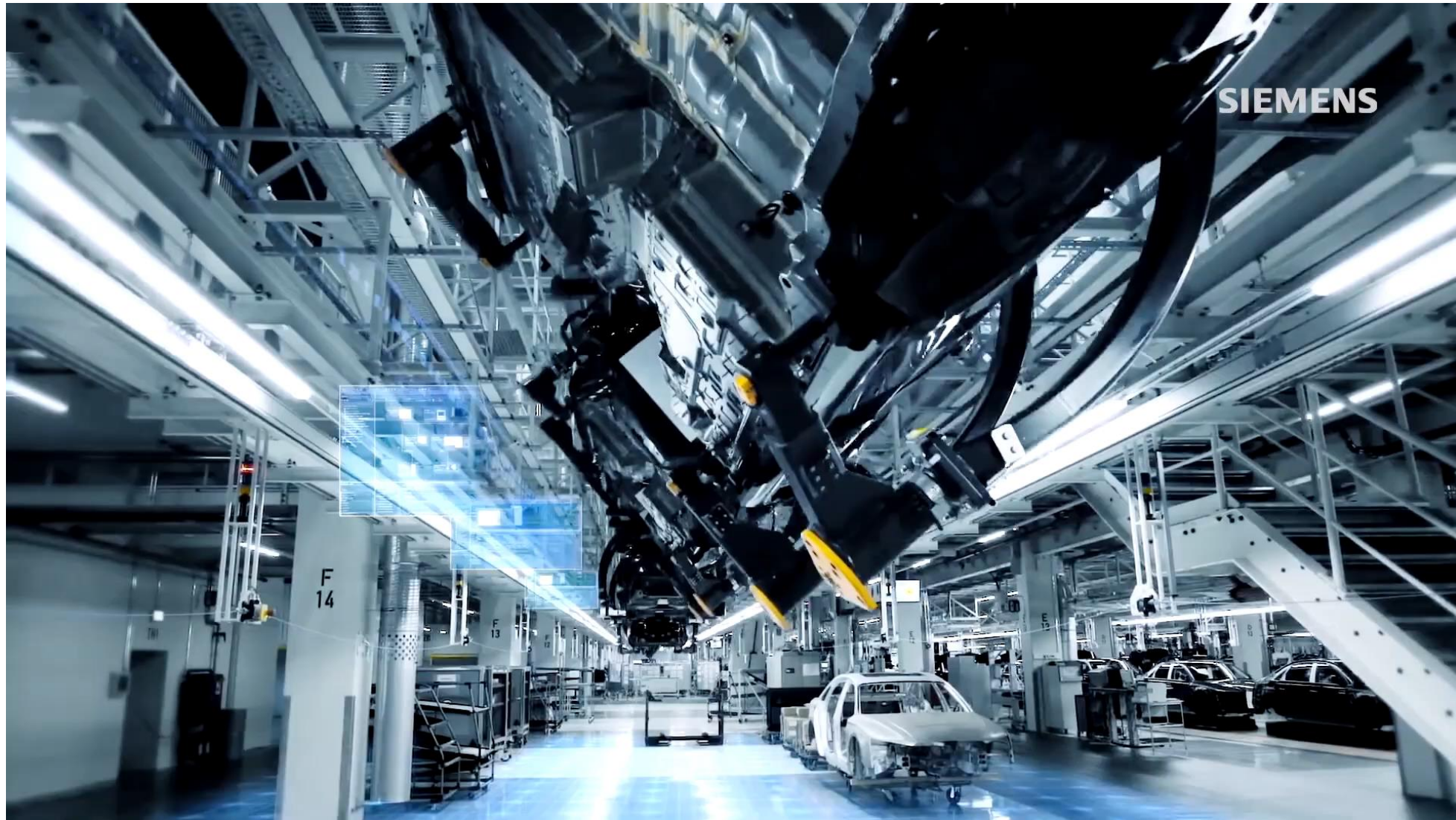
Key Technologies for IIoT: Cyber-Physical Systems

- The interconnection among the different objects such as sensor-enabled machines, processes, storage, people, and the network edge are connected with the help of the Internet.
- The characteristics of the interconnected elements are categorized as follows,



Cyber-Physical Systems

Self-Reading



Cyber-Physical Systems (CPS)

Self-Reading

- **A CPS system possesses essential characteristics: Sequential input-output communication among multiple processes and real-time feedback control based on sensor data.**
- **The computational resources, communication radio, sensor modules, and high-level programming facilitate the classification of mobile devices as mobile CPS.**
- **Mobile CPS has gained much popularity due to the following features:**



Several input-output devices are interconnected and communicate among themselves with the help of the Internet.



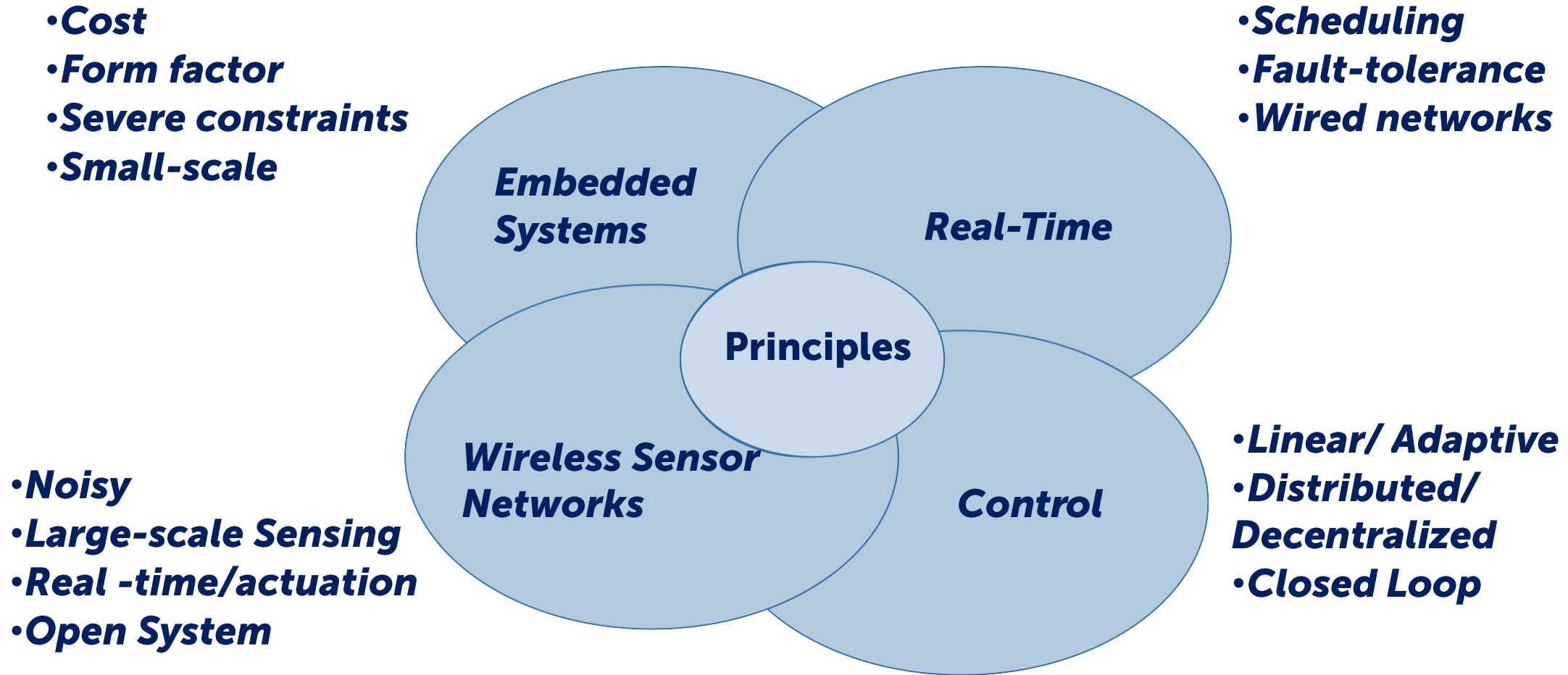
Presence of significant processing devices such as local storage in a smart phone and feedback-based control systems.



Programming languages and automation logic encourage faster development of mobile CPS.

- **Mobile CPS has a wide range of applications, which are as follows:**
 - **Mobile robot communication**
 - **Intelligent transportation system**
 - **Smart technologies: Smart cities, smart healthcare systems, and environmental monitoring for enhanced safety and security and reduced energy consumption.**

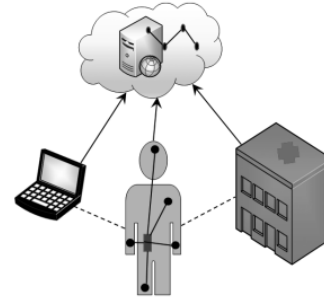
Cyber-Physical Systems: Factors & Constraints

Self-Reading

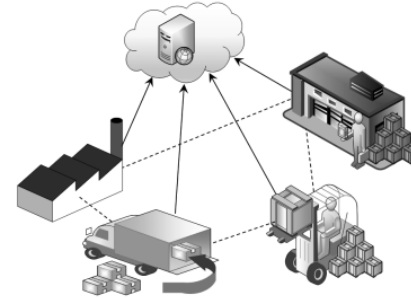
Basic of Industrial IoT

Application of CPS in IIoT

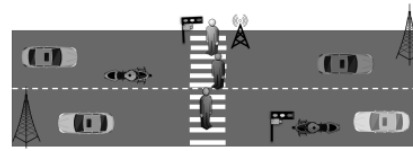
IIoT has a wide range of applications across various manufacturing industries. The application of Computer-Integrated Manufacturing (CIM) systems on the shop floor of industries has led to the development of Product Data Management (PDM)



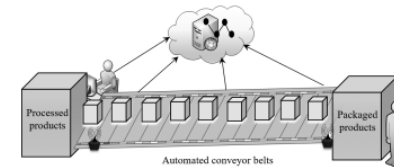
(a) Smart healthcare.



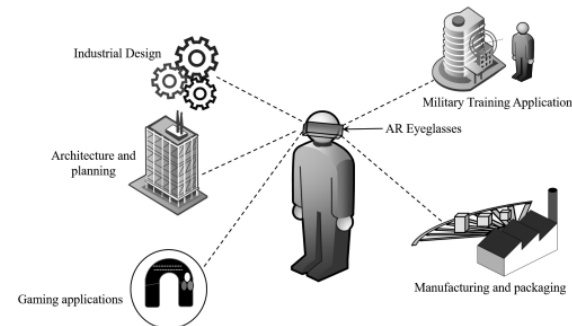
(b) Smart supplychain.



(c) Smart transportation.



(d) Automated packaging system.



(e) Augmented reality and virtual reality applications.

IIoT - Business Challenges

Self Reading

The major challenges faced in the development of IIoT-based business models are



As the physical worlds and virtual worlds combine at a large scale, it is quite difficult to preserve the security and handle data privacy issues.



The organizations require improved security frameworks, which provide authentication at the device-level for system-wide incidence response models.



With the lack of interoperability among the existing systems, the complexity and deployment cost of the systems increases significantly.



To share data seamlessly between the machines and other physical systems (M2M communication) will be quite expensive for the existing technologies.

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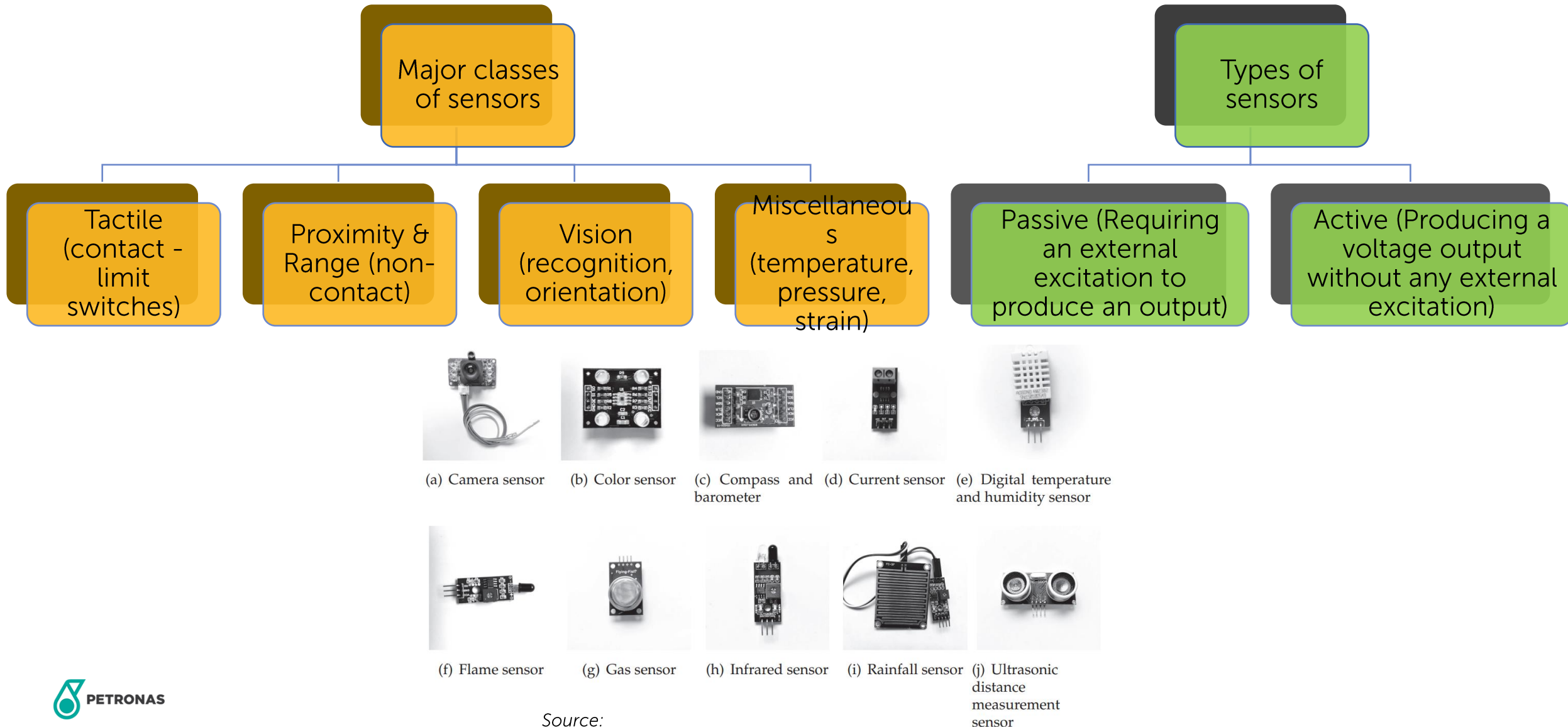
Industrial Sensing & Actuation

- Purpose: monitor, measure, and control various processes, machinery, and systems
- Key Component of Industrial Sensing & Actuation
 1. Sensors : Temperature , Motion sensors , Light sensors , Proximity sensors , Level Sensors , Flow sensors , Accelerometer, Chemical sensors etc
 2. Actuators: Valve, solenoids, motor etc.



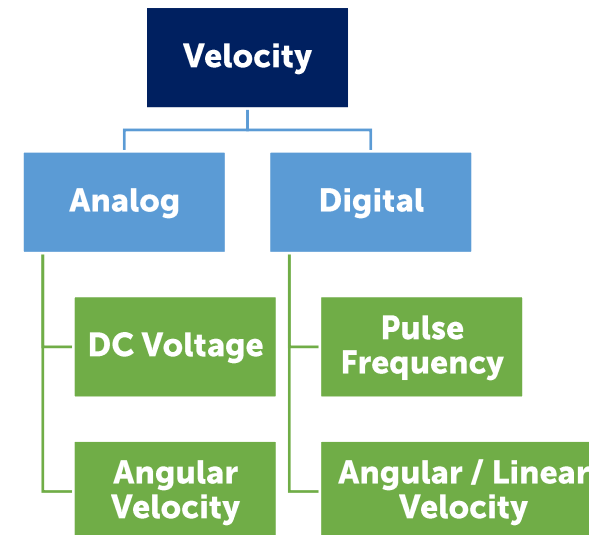
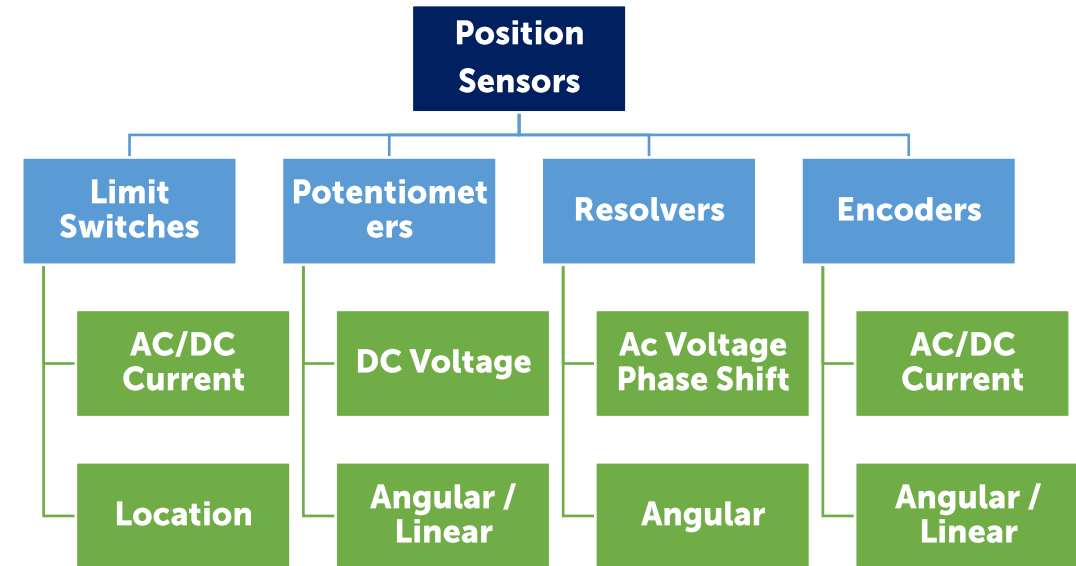
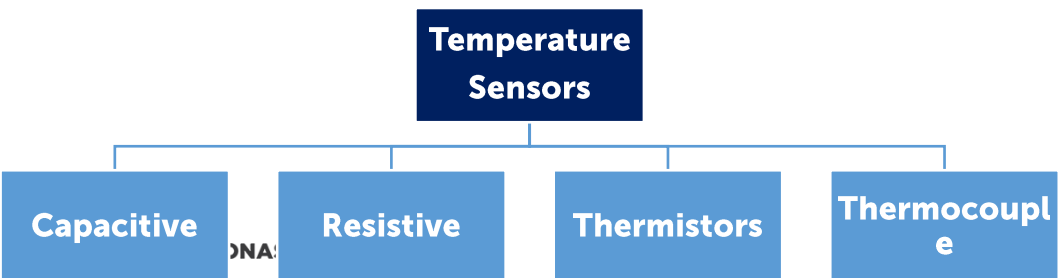
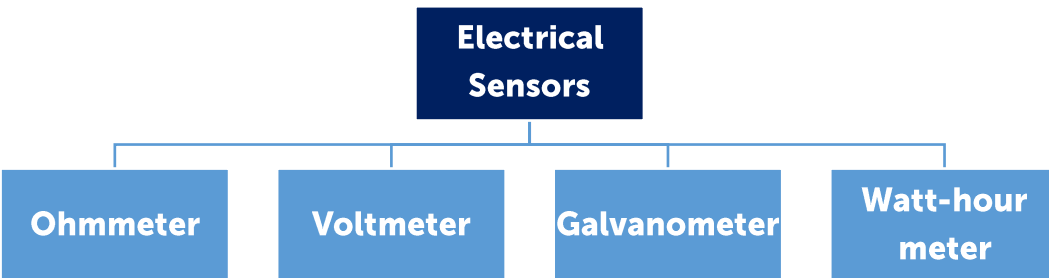
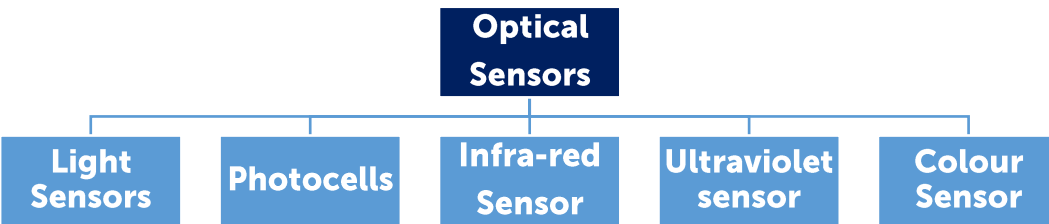
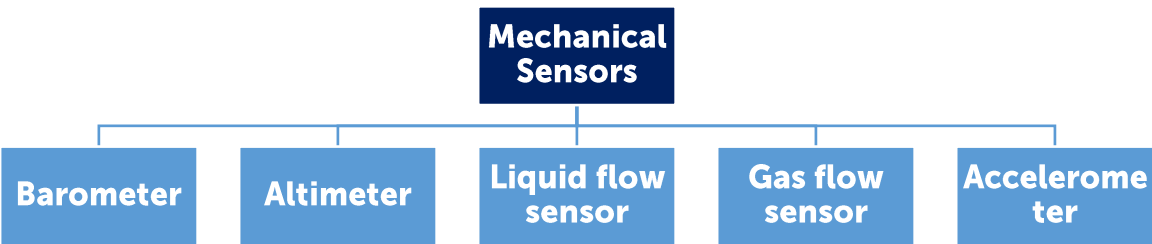
Industrial Sensing & Actuation

Classes of Sensors



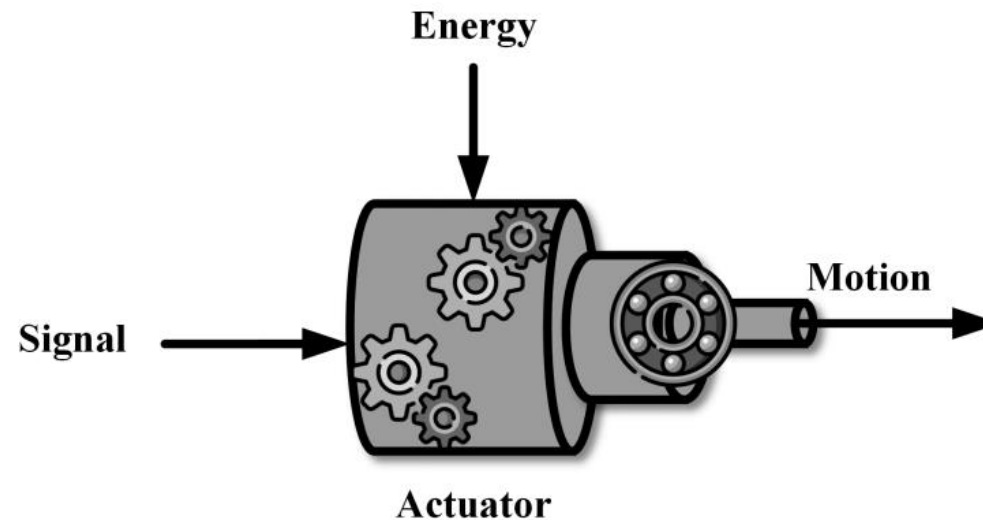
Types of Sensors

Self Reading

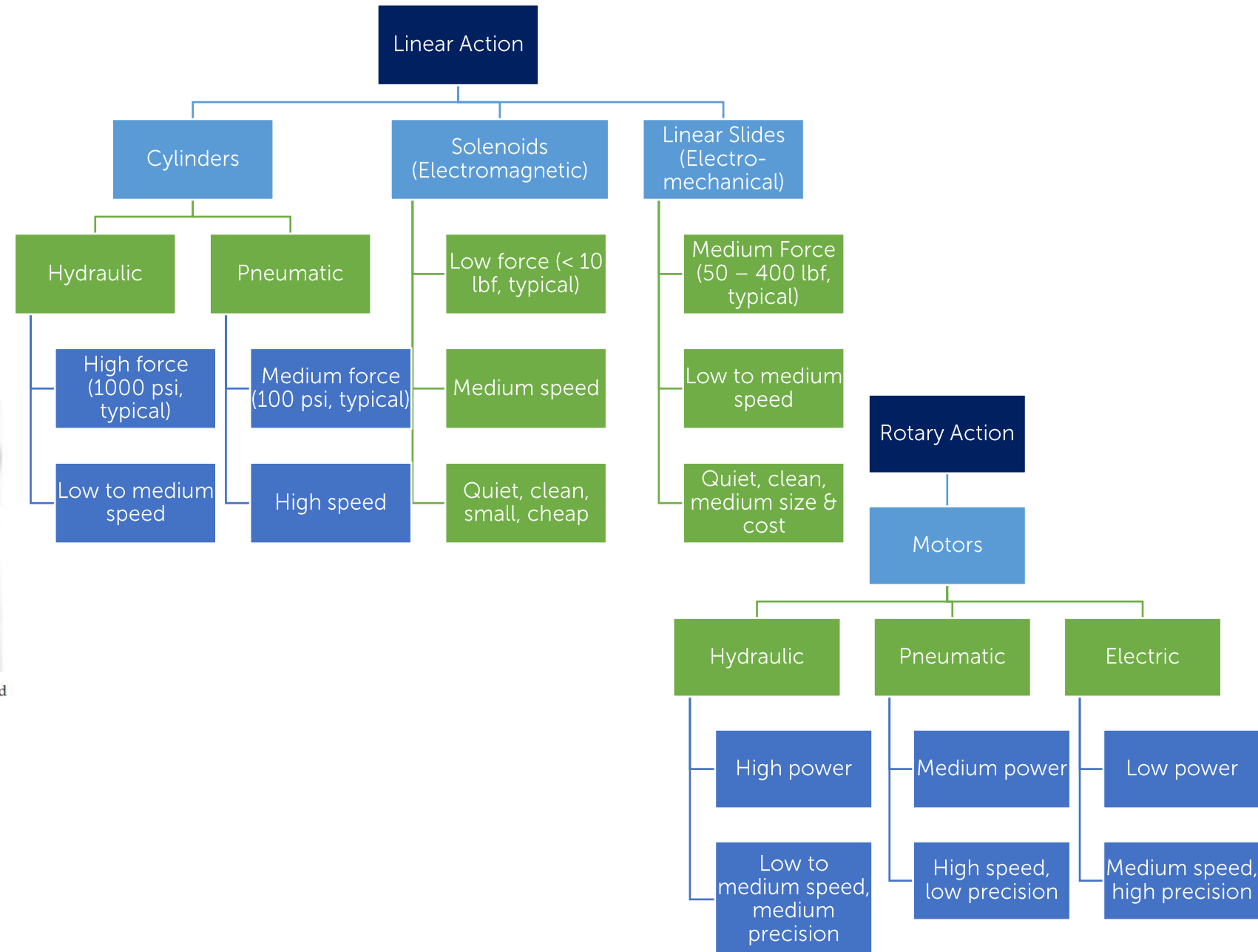


Actuators

- An actuator is a device that converts input signals into physical outputs or mechanical motion.
- The actuator input signals are in the form of electrical signals or thermal set-points.
- Typically, in scenarios such as IoT and IIoT, a processor/controller sends electrical signal patterns to the actuators connected to them.
- Some common examples of actuators include solenoid valves, motors, and pumps.



Types of Actuators

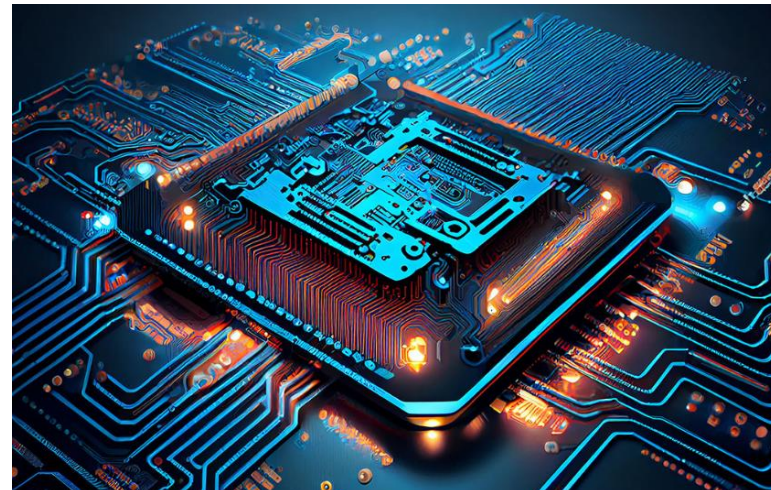
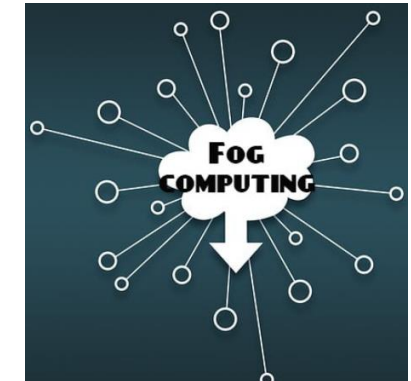
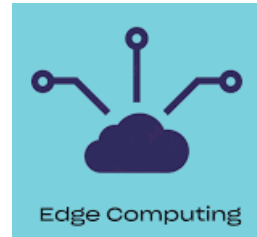


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- IIoT Connectivity

IIoT Compute

- IIoT compute refers to the computing capabilities that enable data processing, analysis, and decision-making within IIoT system which involves a combination of hardware and software technologies that allow industrial systems to process data.
- Key aspects of IIOT Compute :
 - Edge Computing
 - Cloud Computing
 - Fog Computing
 - Data Processing & Analytics
 - AI & Machine Learning
 - Security



IIoT Compute

Cloud vs Fog Vs Edge Computing

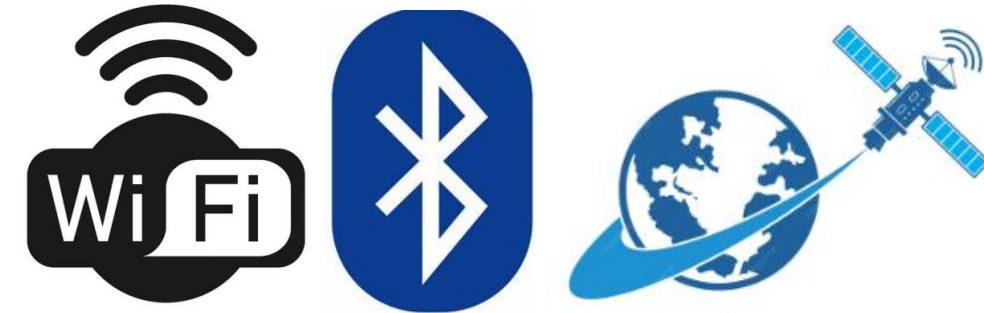
Aspect	Cloud Computing	Fog Computing	Edge Computing
Data Processing	Centralized in the cloud	Distributed (near the edge)	Local, on the device
Latency	High (due to distance)	Moderate (closer to data)	Very low (instantaneous)
Use Cases	Large-scale analytics, data storage	Real-time analytics, IoT systems	Critical, real-time tasks (e.g., autonomous systems)
Scalability	Highly scalable	Moderate scalability	Limited by device resources
Complexity	Centralized, easier management	Intermediate layer, more complex	Decentralized, requires management of multiple devices

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IIoT Connectivity

- IIoT connectivity varies depending on the system requirement e.g. speed, range, power consumption, types of data.
- This range, data transmission is very crucial in the closed-loop control system to take proper control actions
- Common methods of IIoT connectivity includes :
 - Wired – Ethernet , Fieldbus
 - Wireless
 - Wi-Fi
 - Cellular Connectivity (2G/3G/4G/5G)
 - Bluetooth
 - LoRaWAN (Long Range Wide Area Network)
 - Satellite



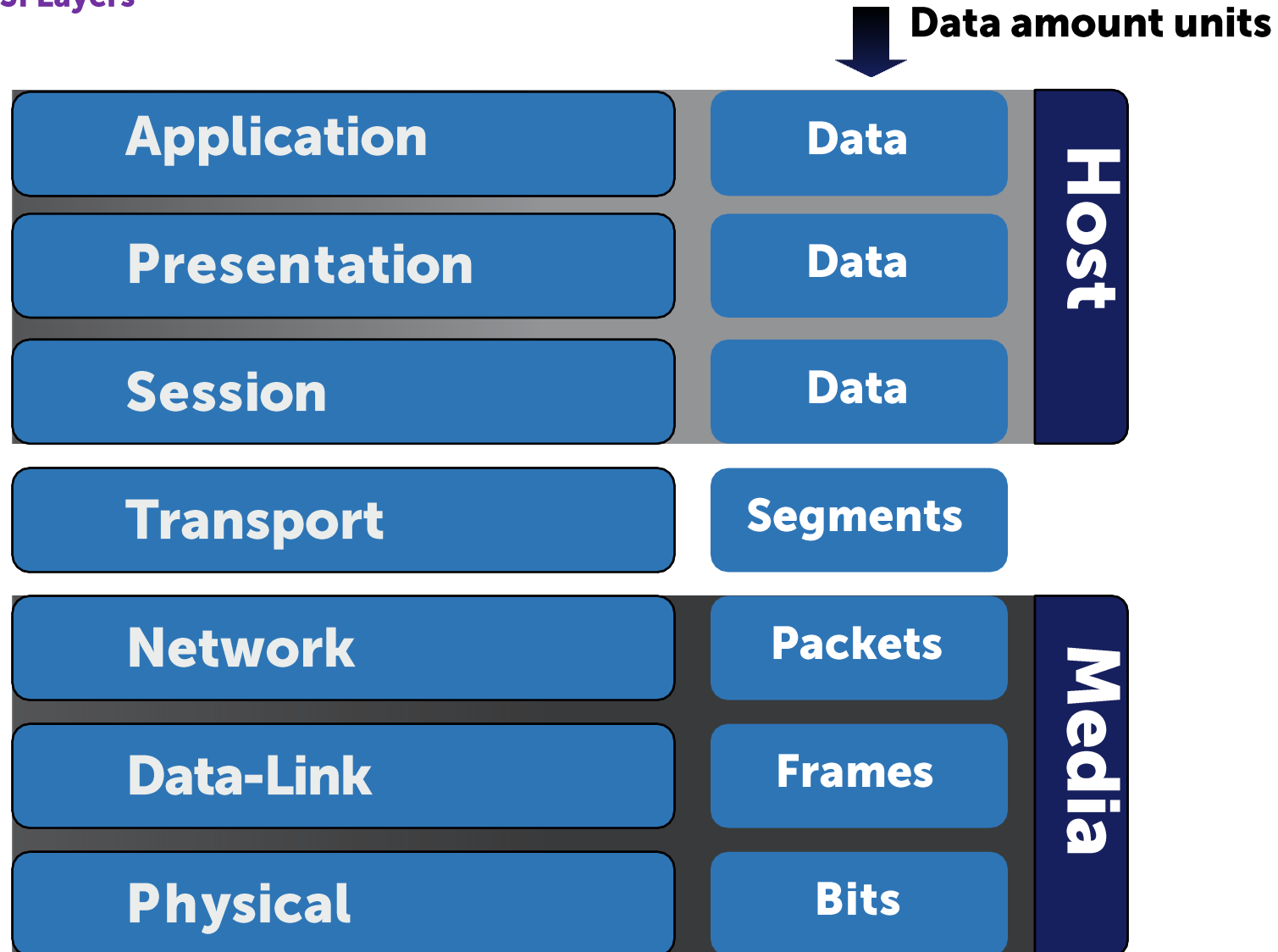
IIoT Connectivity

IIoT Standards

								
Standard	ZigBee (WPAN)	Low Power Wi-Fi (WLAN)	6LoWPAN (LPWAN)	LoRaWAN (LPWAN)	NB-IoT (LPWAN - cellular)	LTE-M (LPWAN - cellular)	5G (cellular)	Wi-Sun (WNAN)
Nominal range	10 -100 m	70 m - 225 m	25 - 50 m	2 - 15 Km	1 - 15 Km	1 - 11 Km	up to 100 km	5 - 10 km
Max Data Rate (Kbit/s)	250 Kbps	15 Mbps	250 Kbps	50 Kbps	250 Kbps	1 Mbps	599 Mbps	300 Kbps
Power consumption	 Medium	 Low to medium	 Low	 Low to medium	 Low	 Low	 Low to medium	 Medium to high

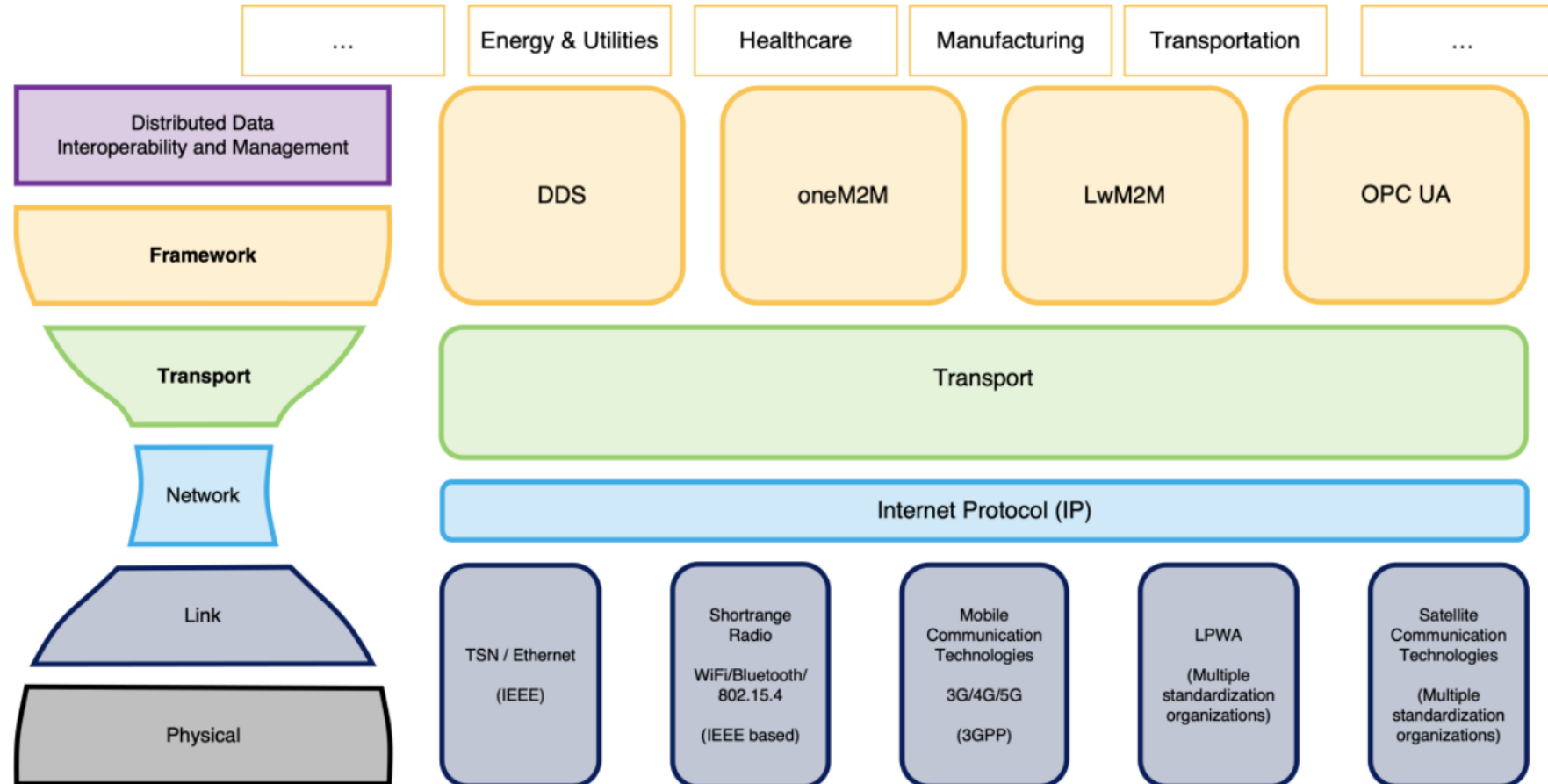
IIoT Connectivity

Communication: OSI Layers



IIoT Connectivity

Communication: OSI Layers



Source: <https://www.iotworldtoday.com/iiot/iic-s-industrial-internet-connectivity-framework-and-why-you-should-care>

